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# **NUMERICAL METHODS**

## **CPE 312**

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*Lecturer*



## RECALL:

1. It is a mathematical procedure that uses an initial value to generate a sequence of improving approximate solutions for a class of problems.



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# Indirect or Iterative Methods



2. These techniques are based on making two initial guesses that bracket the root- that is, are on either side of the root.



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# Bracketing Method



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3. A simple method for obtaining an estimate of the root of the equation  $f(x) = 0$  is to make a plot of a function and observe where it crosses the x-axis.



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# Graphical Method



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# **INCREMENTAL-SEARCH METHOD**





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## **Learning Objectives**

- \*Define Incremental-Search Method**
- \*Determine and perform the algorithm of Incremental-Search Method**
  - \*Solve functions using Incremental-Search Method**



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- Incremental search method uses a small increment to search for the root in a step-by-step manner.
- If  $f(x_1)$  and  $f(x_2)$  have opposite signs, then there is at least one root in the intervals  $(x_1, x_2)$



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- The zeros of  $f(x)$  can be detected by evaluating the function at the interval  $\Delta x$  and look for a change in sign.



## ***Algorithm of Incremental-Search Method:***

1. Select the initial value of  $x_1$  and  $\Delta x$ .
2. Determine the values of  $f(x_1)$  and  $f(x_n)$ ;

$$x_n = x_1 + \Delta x.$$

3.  $f(x_1) \cdot f(x_n) > 0$     continue  
    $f(x_1) \cdot f(x_n) < 0$     revert



4. If there is a sign changed, revert back to  $x_1$  and reduce  $\Delta x$  to smaller increment.
5. Determine  $f(x_1)$  and  $f(x_n)$  progressively until another sign change.
6. Repeat steps until  $f(x_n) \approx 0$ .  
(stopping criterion 0.000...)



## ***Illustrative Example # 1:***

Use the incremental-search method to determine the root of  $f(x) = e^{-x} - x$ . Use  $x_1 = 0$  and  $\Delta x = 0.5$

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[illegible]



| Iterations | $x_1$ | $\Delta x$ | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|------------|-------|------------|-------|----------|----------|-----------------------|----------|
| 1          | 0     | 0.5        | 0.5   | 1        | 0.10653  | $> 0$                 | continue |
| 2          | 0.5   | 0.5        | 1     | 0.10653  | -0.63212 | $< 0$                 | revert   |
|            |       |            |       |          |          |                       |          |
|            |       |            |       |          |          |                       |          |
|            |       |            |       |          |          |                       |          |
|            |       |            |       |          |          |                       |          |
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| 3          | 0.5   | 0.1        | 0.6   | 0.10653  |          |                       |          |
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| 3          | 0.5   | 0.1        | 0.6   | 0.10653  | -0.05119 |                       |          |
|            |       |            |       |          |          |                       |          |
|            |       |            |       |          |          |                       |          |
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| 4          | 0.5   | 0.05       | 0.55  | 0.10653  | 0.02695  |                       |          |
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| 6          | 0.55  | 0.01       | 0.56  | 0.02695  | 0.01121  | $> 0$                 | continue |
| 7          | 0.56  | 0.01       | 0.57  | 0.01121  |          |                       |          |



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| 6          | 0.55  | 0.01       | 0.56  | 0.02695  | 0.01121  | $> 0$                 | continue |
| 7          | 0.56  | 0.01       | 0.57  | 0.01121  | -0.00447 |                       |          |



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| Iterations | $x_1$ | $x$    | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|------------|-------|--------|-------|----------|----------|-----------------------|----------|
| 8          | 0.56  | 0.0001 | 0.561 | 0.01121  | 0.00964  | $> 0$                 | continue |
| 9          | 0.561 | 0.0001 | 0.562 | 0.00964  | 0.00807  | $> 0$                 | continue |
| 10         | 0.562 | 0.0001 | 0.563 | 0.00807  | 0.0065   |                       |          |
|            |       |        |       |          |          |                       |          |
|            |       |        |       |          |          |                       |          |
|            |       |        |       |          |          |                       |          |
|            |       |        |       |          |          |                       |          |
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| 8          | 0.56  | 0.0001 | 0.561 | 0.01121  | 0.00964  | $> 0$                 | continue |
| 9          | 0.561 | 0.0001 | 0.562 | 0.00964  | 0.00807  | $> 0$                 | continue |
| 10         | 0.562 | 0.0001 | 0.563 | 0.00807  | 0.0065   | $> 0$                 | continue |
|            |       |        |       |          |          |                       |          |
|            |       |        |       |          |          |                       |          |
|            |       |        |       |          |          |                       |          |
|            |       |        |       |          |          |                       |          |
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| 8               | 0.56  | 0.001 | 0.561 | 0.01121  | 0.00964  | $> 0$                 | continue |
| 9               | 0.561 | 0.001 | 0.562 | 0.00964  | 0.00807  | $> 0$                 | continue |
| 10              | 0.562 | 0.001 | 0.563 | 0.00807  | 0.0065   | $> 0$                 | continue |
| 11              | 0.563 | 0.001 | 0.564 | 0.0065   |          |                       |          |
|                 |       |       |       |          |          |                       |          |
|                 |       |       |       |          |          |                       |          |
|                 |       |       |       |          |          |                       |          |
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|            |       |       |       |          |          |                       |          |
|            |       |       |       |          |          |                       |          |
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| Iteration<br>ns | $x_1$ | $x$   | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|-----------------|-------|-------|-------|----------|----------|-----------------------|----------|
| 8               | 0.56  | 0.001 | 0.561 | 0.01121  | 0.00964  | $> 0$                 | continue |
| 9               | 0.561 | 0.001 | 0.562 | 0.00964  | 0.00807  | $> 0$                 | continue |
| 10              | 0.562 | 0.001 | 0.563 | 0.00807  | 0.0065   | $> 0$                 | continue |
| 11              | 0.563 | 0.001 | 0.564 | 0.0065   | 0.00493  | $> 0$                 | continue |
| 12              | 0.564 | 0.001 | 0.565 | 0.00493  |          |                       |          |
|                 |       |       |       |          |          |                       |          |
|                 |       |       |       |          |          |                       |          |
|                 |       |       |       |          |          |                       |          |



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| Iteratio<br>ns | $x_1$ | $x$   | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|----------------|-------|-------|-------|----------|----------|-----------------------|----------|
| 8              | 0.56  | 0.001 | 0.561 | 0.01121  | 0.00964  | $> 0$                 | continue |
| 9              | 0.561 | 0.001 | 0.562 | 0.00964  | 0.00807  | $> 0$                 | continue |
| 10             | 0.562 | 0.001 | 0.563 | 0.00807  | 0.0065   | $> 0$                 | continue |
| 11             | 0.563 | 0.001 | 0.564 | 0.0065   | 0.00493  | $> 0$                 | continue |
| 12             | 0.564 | 0.001 | 0.565 | 0.00493  | 0.00336  |                       |          |
|                |       |       |       |          |          |                       |          |
|                |       |       |       |          |          |                       |          |
|                |       |       |       |          |          |                       |          |



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| Iteration<br>ns | $x_1$ | $x$   | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
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| 8               | 0.56  | 0.001 | 0.561 | 0.01121  | 0.00964  | $> 0$                 | continue |
| 9               | 0.561 | 0.001 | 0.562 | 0.00964  | 0.00807  | $> 0$                 | continue |
| 10              | 0.562 | 0.001 | 0.563 | 0.00807  | 0.0065   | $> 0$                 | continue |
| 11              | 0.563 | 0.001 | 0.564 | 0.0065   | 0.00493  | $> 0$                 | continue |
| 12              | 0.564 | 0.001 | 0.565 | 0.00493  | 0.00336  | $> 0$                 | continue |
|                 |       |       |       |          |          |                       |          |
|                 |       |       |       |          |          |                       |          |
|                 |       |       |       |          |          |                       |          |



# PAMANTASAN NG LUNGSOD NG SAN PABLO

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| Iterations | $x_1$ | $x$   | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|------------|-------|-------|-------|----------|----------|-----------------------|----------|
| 8          | 0.56  | 0.001 | 0.561 | 0.01121  | 0.00964  | $> 0$                 | continue |
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| 10         | 0.562 | 0.001 | 0.563 | 0.00807  | 0.0065   | $> 0$                 | continue |
| 11         | 0.563 | 0.001 | 0.564 | 0.0065   | 0.00493  | $> 0$                 | continue |
| 12         | 0.564 | 0.001 | 0.565 | 0.00493  | 0.00336  | $> 0$                 | continue |
| 13         | 0.565 | 0.001 | 0.566 | 0.00336  |          |                       |          |
|            |       |       |       |          |          |                       |          |
|            |       |       |       |          |          |                       |          |



# PAMANTASAN NG LUNGSOD NG SAN PABLO

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| Iteratio<br>ns | $x_1$ | $x$   | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|----------------|-------|-------|-------|----------|----------|-----------------------|----------|
| 8              | 0.56  | 0.001 | 0.561 | 0.01121  | 0.00964  | $> 0$                 | continue |
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| 10             | 0.562 | 0.001 | 0.563 | 0.00807  | 0.0065   | $> 0$                 | continue |
| 11             | 0.563 | 0.001 | 0.564 | 0.0065   | 0.00493  | $> 0$                 | continue |
| 12             | 0.564 | 0.001 | 0.565 | 0.00493  | 0.00336  | $> 0$                 | continue |
| 13             | 0.565 | 0.001 | 0.566 | 0.00336  | 0.00179  |                       |          |
|                |       |       |       |          |          |                       |          |
|                |       |       |       |          |          |                       |          |



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| Iterations | $x_1$ | $x$   | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|------------|-------|-------|-------|----------|----------|-----------------------|----------|
| 8          | 0.56  | 0.001 | 0.561 | 0.01121  | 0.00964  | $> 0$                 | continue |
| 9          | 0.561 | 0.001 | 0.562 | 0.00964  | 0.00807  | $> 0$                 | continue |
| 10         | 0.562 | 0.001 | 0.563 | 0.00807  | 0.0065   | $> 0$                 | continue |
| 11         | 0.563 | 0.001 | 0.564 | 0.0065   | 0.00493  | $> 0$                 | continue |
| 12         | 0.564 | 0.001 | 0.565 | 0.00493  | 0.00336  | $> 0$                 | continue |
| 13         | 0.565 | 0.001 | 0.566 | 0.00336  | 0.00179  | $> 0$                 | continue |
|            |       |       |       |          |          |                       |          |
|            |       |       |       |          |          |                       |          |



# PAMANTASAN NG LUNGSOD NG SAN PABLO

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| Iterations | $x_1$ | $x$   | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|------------|-------|-------|-------|----------|----------|-----------------------|----------|
| 8          | 0.56  | 0.001 | 0.561 | 0.01121  | 0.00964  | $> 0$                 | continue |
| 9          | 0.561 | 0.001 | 0.562 | 0.00964  | 0.00807  | $> 0$                 | continue |
| 10         | 0.562 | 0.001 | 0.563 | 0.00807  | 0.0065   | $> 0$                 | continue |
| 11         | 0.563 | 0.001 | 0.564 | 0.0065   | 0.00493  | $> 0$                 | continue |
| 12         | 0.564 | 0.001 | 0.565 | 0.00493  | 0.00336  | $> 0$                 | continue |
| 13         | 0.565 | 0.001 | 0.566 | 0.00336  | 0.00179  | $> 0$                 | continue |
| 14         | 0.566 | 0.001 | 0.567 | 0.00179  |          |                       |          |
|            |       |       |       |          |          |                       |          |



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| Iteration<br>ns | $x_1$ | $x$   | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|-----------------|-------|-------|-------|----------|----------|-----------------------|----------|
| 8               | 0.56  | 0.001 | 0.561 | 0.01121  | 0.00964  | $> 0$                 | continue |
| 9               | 0.561 | 0.001 | 0.562 | 0.00964  | 0.00807  | $> 0$                 | continue |
| 10              | 0.562 | 0.001 | 0.563 | 0.00807  | 0.0065   | $> 0$                 | continue |
| 11              | 0.563 | 0.001 | 0.564 | 0.0065   | 0.00493  | $> 0$                 | continue |
| 12              | 0.564 | 0.001 | 0.565 | 0.00493  | 0.00336  | $> 0$                 | continue |
| 13              | 0.565 | 0.001 | 0.566 | 0.00336  | 0.00179  | $> 0$                 | continue |
| 14              | 0.566 | 0.001 | 0.567 | 0.00179  | 0.00022  |                       |          |
|                 |       |       |       |          |          |                       |          |





# PAMANTASAN NG LUNGSOD NG SAN PABLO

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| Iteration<br>ns | $x_1$ | $x$   | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|-----------------|-------|-------|-------|----------|----------|-----------------------|----------|
| 8               | 0.56  | 0.001 | 0.561 | 0.01121  | 0.00964  | $> 0$                 | continue |
| 9               | 0.561 | 0.001 | 0.562 | 0.00964  | 0.00807  | $> 0$                 | continue |
| 10              | 0.562 | 0.001 | 0.563 | 0.00807  | 0.0065   | $> 0$                 | continue |
| 11              | 0.563 | 0.001 | 0.564 | 0.0065   | 0.00493  | $> 0$                 | continue |
| 12              | 0.564 | 0.001 | 0.565 | 0.00493  | 0.00336  | $> 0$                 | continue |
| 13              | 0.565 | 0.001 | 0.566 | 0.00336  | 0.00179  | $> 0$                 | continue |
| 14              | 0.566 | 0.001 | 0.567 | 0.00179  | 0.00022  | $> 0$                 | continue |
|                 |       |       |       |          |          |                       |          |



# PAMANTASAN NG LUNGSOD NG SAN PABLO

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| Iterations | $x_1$ | $x$   | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|------------|-------|-------|-------|----------|----------|-----------------------|----------|
| 8          | 0.56  | 0.001 | 0.561 | 0.01121  | 0.00964  | $> 0$                 | continue |
| 9          | 0.561 | 0.001 | 0.562 | 0.00964  | 0.00807  | $> 0$                 | continue |
| 10         | 0.562 | 0.001 | 0.563 | 0.00807  | 0.0065   | $> 0$                 | continue |
| 11         | 0.563 | 0.001 | 0.564 | 0.0065   | 0.00493  | $> 0$                 | continue |
| 12         | 0.564 | 0.001 | 0.565 | 0.00493  | 0.00336  | $> 0$                 | continue |
| 13         | 0.565 | 0.001 | 0.566 | 0.00336  | 0.00179  | $> 0$                 | continue |
| 14         | 0.566 | 0.001 | 0.567 | 0.00179  | 0.00022  | $> 0$                 | continue |



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The approximate root of  
 $f(x) = e^{-x} - x$  is 0.567.



## ***Illustrative Example # 2:***

Use the incremental-search method to determine the root of  $f(x) = 4x^3 + 3x^2 + 2x + 1$ .  
Use  $x_1 = -1$  and  $\Delta x = 0.4$ .

[illegible]

[illegible]

[illegible]

[illegible]



[illegible]

[illegible]

[illegible]







| Iterations | $x_1$ | $\Delta x$ | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|------------|-------|------------|-------|----------|----------|-----------------------|----------|
| 1          | -1    | 0.4        | -0.6  | -2       | 0.016    | $< 0$                 | revert   |
| 2          | -1    | 0.2        | -0.8  | -2       | -0.728   | $> 0$                 | continue |
| 3          | -0.8  | 0.2        | -0.6  | -0.728   | 0.016    | $< 0$                 | revert   |
|            |       |            |       |          |          |                       |          |
|            |       |            |       |          |          |                       |          |
|            |       |            |       |          |          |                       |          |
|            |       |            |       |          |          |                       |          |
|            |       |            |       |          |          |                       |          |



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| Iteratio<br>ns | $x_1$ | $\triangle x$ | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|----------------|-------|---------------|-------|----------|----------|-----------------------|----------|
| 1              | -1    | 0.4           | -0.6  | -2       | 0.016    | $< 0$                 | revert   |
| 2              | -1    | 0.2           | -0.8  | -2       | -0.728   | $> 0$                 | continue |
| 3              | -0.8  | 0.2           | -0.6  | -0.728   | 0.016    | $< 0$                 | revert   |
| 4              | -0.8  | 0.1           | -0.7  | -0.728   |          |                       |          |
|                |       |               |       |          |          |                       |          |
|                |       |               |       |          |          |                       |          |
|                |       |               |       |          |          |                       |          |
|                |       |               |       |          |          |                       |          |



# PAMANTASAN NG LUNGSOD NG SAN PABLO

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| Iteratio<br>ns | $x_1$ | $\triangle x$ | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|----------------|-------|---------------|-------|----------|----------|-----------------------|----------|
| 1              | -1    | 0.4           | -0.6  | -2       | 0.016    | $< 0$                 | revert   |
| 2              | -1    | 0.2           | -0.8  | -2       | -0.728   | $> 0$                 | continue |
| 3              | -0.8  | 0.2           | -0.6  | -0.728   | 0.016    | $< 0$                 | revert   |
| 4              | -0.8  | 0.1           | -0.7  | -0.728   | -0.302   |                       |          |
|                |       |               |       |          |          |                       |          |
|                |       |               |       |          |          |                       |          |
|                |       |               |       |          |          |                       |          |
|                |       |               |       |          |          |                       |          |
|                |       |               |       |          |          |                       |          |





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| Iteratio<br>ns | $x_1$ | $\triangle x$ | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|----------------|-------|---------------|-------|----------|----------|-----------------------|----------|
| 1              | -1    | 0.4           | -0.6  | -2       | 0.016    | $< 0$                 | revert   |
| 2              | -1    | 0.2           | -0.8  | -2       | -0.728   | $> 0$                 | continue |
| 3              | -0.8  | 0.2           | -0.6  | -0.728   | 0.016    | $< 0$                 | revert   |
| 4              | -0.8  | 0.1           | -0.7  | -0.728   | -0.302   | $> 0$                 | continue |
|                |       |               |       |          |          |                       |          |
|                |       |               |       |          |          |                       |          |
|                |       |               |       |          |          |                       |          |
|                |       |               |       |          |          |                       |          |
|                |       |               |       |          |          |                       |          |



# PAMANTASAN NG LUNGSOD NG SAN PABLO

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| Iteratio<br>ns | $x_1$ | $\triangle x$ | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|----------------|-------|---------------|-------|----------|----------|-----------------------|----------|
| 1              | -1    | 0.4           | -0.6  | -2       | 0.016    | $< 0$                 | revert   |
| 2              | -1    | 0.2           | -0.8  | -2       | -0.728   | $> 0$                 | continue |
| 3              | -0.8  | 0.2           | -0.6  | -0.728   | 0.016    | $< 0$                 | revert   |
| 4              | -0.8  | 0.1           | -0.7  | -0.728   | -0.302   | $> 0$                 | continue |
| 5              | -0.7  | 0.1           | -0.6  | -0.302   |          |                       |          |
|                |       |               |       |          |          |                       |          |
|                |       |               |       |          |          |                       |          |
|                |       |               |       |          |          |                       |          |



# PAMANTASAN NG LUNGSOD NG SAN PABLO

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| Iteratio<br>ns | $x_1$ | $\triangle x$ | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|----------------|-------|---------------|-------|----------|----------|-----------------------|----------|
| 1              | -1    | 0.4           | -0.6  | -2       | 0.016    | $< 0$                 | revert   |
| 2              | -1    | 0.2           | -0.8  | -2       | -0.728   | $> 0$                 | continue |
| 3              | -0.8  | 0.2           | -0.6  | -0.728   | 0.016    | $< 0$                 | revert   |
| 4              | -0.8  | 0.1           | -0.7  | -0.728   | -0.302   | $> 0$                 | continue |
| 5              | -0.7  | 0.1           | -0.6  | -0.302   | 0.016    |                       |          |
|                |       |               |       |          |          |                       |          |
|                |       |               |       |          |          |                       |          |
|                |       |               |       |          |          |                       |          |



# PAMANTASAN NG LUNGSOD NG SAN PABLO

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| Iteration<br>ns | $x_1$ | $\Delta x$ | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|-----------------|-------|------------|-------|----------|----------|-----------------------|----------|
| 1               | -1    | 0.4        | -0.6  | -2       | 0.016    | $< 0$                 | revert   |
| 2               | -1    | 0.2        | -0.8  | -2       | -0.728   | $> 0$                 | continue |
| 3               | -0.8  | 0.2        | -0.6  | -0.728   | 0.016    | $< 0$                 | revert   |
| 4               | -0.8  | 0.1        | -0.7  | -0.728   | -0.302   | $> 0$                 | continue |
| 5               | -0.7  | 0.1        | -0.6  | -0.302   | 0.016    | $< 0$                 | revert   |
|                 |       |            |       |          |          |                       |          |
|                 |       |            |       |          |          |                       |          |
|                 |       |            |       |          |          |                       |          |



# PAMANTASAN NG LUNGSOD NG SAN PABLO

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| Iteratio<br>ns | $x_1$ | $\triangle x$ | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|----------------|-------|---------------|-------|----------|----------|-----------------------|----------|
| 1              | -1    | 0.4           | -0.6  | -2       | 0.016    | $< 0$                 | revert   |
| 2              | -1    | 0.2           | -0.8  | -2       | -0.728   | $> 0$                 | continue |
| 3              | -0.8  | 0.2           | -0.6  | -0.728   | 0.016    | $< 0$                 | revert   |
| 4              | -0.8  | 0.1           | -0.7  | -0.728   | -0.302   | $> 0$                 | continue |
| 5              | -0.7  | 0.1           | -0.6  | -0.302   | 0.016    | $< 0$                 | revert   |
| 6              | -0.7  | 0.05          | -0.65 | -0.302   |          |                       |          |
|                |       |               |       |          |          |                       |          |
|                |       |               |       |          |          |                       |          |



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| Iteratio<br>ns | $x_1$ | $\triangle x$ | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|----------------|-------|---------------|-------|----------|----------|-----------------------|----------|
| 1              | -1    | 0.4           | -0.6  | -2       | 0.016    | $< 0$                 | revert   |
| 2              | -1    | 0.2           | -0.8  | -2       | -0.728   | $> 0$                 | continue |
| 3              | -0.8  | 0.2           | -0.6  | -0.728   | 0.016    | $< 0$                 | revert   |
| 4              | -0.8  | 0.1           | -0.7  | -0.728   | -0.302   | $> 0$                 | continue |
| 5              | -0.7  | 0.1           | -0.6  | -0.302   | 0.016    | $< 0$                 | revert   |
| 6              | -0.7  | 0.05          | -0.65 | -0.302   | -0.131   |                       |          |
|                |       |               |       |          |          |                       |          |
|                |       |               |       |          |          |                       |          |



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| Iteratio<br>ns | $x_1$ | $\triangle x$ | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|----------------|-------|---------------|-------|----------|----------|-----------------------|----------|
| 1              | -1    | 0.4           | -0.6  | -2       | 0.016    | $< 0$                 | revert   |
| 2              | -1    | 0.2           | -0.8  | -2       | -0.728   | $> 0$                 | continue |
| 3              | -0.8  | 0.2           | -0.6  | -0.728   | 0.016    | $< 0$                 | revert   |
| 4              | -0.8  | 0.1           | -0.7  | -0.728   | -0.302   | $> 0$                 | continue |
| 5              | -0.7  | 0.1           | -0.6  | -0.302   | 0.016    | $< 0$                 | revert   |
| 6              | -0.7  | 0.05          | -0.65 | -0.302   | -0.131   | $> 0$                 | continue |
|                |       |               |       |          |          |                       |          |
|                |       |               |       |          |          |                       |          |



# PAMANTASAN NG LUNGSOD NG SAN PABLO

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| Iterations | $x_1$ | $\Delta x$ | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|------------|-------|------------|-------|----------|----------|-----------------------|----------|
| 1          | -1    | 0.4        | -0.6  | -2       | 0.016    | $< 0$                 | revert   |
| 2          | -1    | 0.2        | -0.8  | -2       | -0.728   | $> 0$                 | continue |
| 3          | -0.8  | 0.2        | -0.6  | -0.728   | 0.016    | $< 0$                 | revert   |
| 4          | -0.8  | 0.1        | -0.7  | -0.728   | -0.302   | $> 0$                 | continue |
| 5          | -0.7  | 0.1        | -0.6  | -0.302   | 0.016    | $< 0$                 | revert   |
| 6          | -0.7  | 0.05       | -0.65 | -0.302   | -0.131   | $> 0$                 | continue |
| 7          | -0.65 | 0.05       | -0.6  | -0.131   |          |                       |          |
|            |       |            |       |          |          |                       |          |





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| Iterations | $x_1$ | $\Delta x$ | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|------------|-------|------------|-------|----------|----------|-----------------------|----------|
| 1          | -1    | 0.4        | -0.6  | -2       | 0.016    | $< 0$                 | revert   |
| 2          | -1    | 0.2        | -0.8  | -2       | -0.728   | $> 0$                 | continue |
| 3          | -0.8  | 0.2        | -0.6  | -0.728   | 0.016    | $< 0$                 | revert   |
| 4          | -0.8  | 0.1        | -0.7  | -0.728   | -0.302   | $> 0$                 | continue |
| 5          | -0.7  | 0.1        | -0.6  | -0.302   | 0.016    | $< 0$                 | revert   |
| 6          | -0.7  | 0.05       | -0.65 | -0.302   | -0.131   | $> 0$                 | continue |
| 7          | -0.65 | 0.05       | -0.6  | -0.131   | 0.016    |                       |          |
|            |       |            |       |          |          |                       |          |



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| Iterations | $x_1$ | $\Delta x$ | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|------------|-------|------------|-------|----------|----------|-----------------------|----------|
| 1          | -1    | 0.4        | -0.6  | -2       | 0.016    | $< 0$                 | revert   |
| 2          | -1    | 0.2        | -0.8  | -2       | -0.728   | $> 0$                 | continue |
| 3          | -0.8  | 0.2        | -0.6  | -0.728   | 0.016    | $< 0$                 | revert   |
| 4          | -0.8  | 0.1        | -0.7  | -0.728   | -0.302   | $> 0$                 | continue |
| 5          | -0.7  | 0.1        | -0.6  | -0.302   | 0.016    | $< 0$                 | revert   |
| 6          | -0.7  | 0.05       | -0.65 | -0.302   | -0.131   | $> 0$                 | continue |
| 7          | -0.65 | 0.05       | -0.6  | -0.131   | 0.016    | $< 0$                 | revert   |
|            |       |            |       |          |          |                       |          |



# PAMANTASAN NG LUNGSOD NG SAN PABLO

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| Iterations | $x_1$ | $\Delta x$ | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|------------|-------|------------|-------|----------|----------|-----------------------|----------|
| 1          | -1    | 0.4        | -0.6  | -2       | 0.016    | $< 0$                 | revert   |
| 2          | -1    | 0.2        | -0.8  | -2       | -0.728   | $> 0$                 | continue |
| 3          | -0.8  | 0.2        | -0.6  | -0.728   | 0.016    | $< 0$                 | revert   |
| 4          | -0.8  | 0.1        | -0.7  | -0.728   | -0.302   | $> 0$                 | continue |
| 5          | -0.7  | 0.1        | -0.6  | -0.302   | 0.016    | $< 0$                 | revert   |
| 6          | -0.7  | 0.05       | -0.65 | -0.302   | -0.131   | $> 0$                 | continue |
| 7          | -0.65 | 0.05       | -0.6  | -0.131   | 0.016    | $< 0$                 | revert   |
| 8          | -0.65 | 0.01       | -0.64 | -0.131   |          |                       |          |



# PAMANTASAN NG LUNGSOD NG SAN PABLO

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| Iteration<br>ns | $x_1$ | $\triangle x$ | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|-----------------|-------|---------------|-------|----------|----------|-----------------------|----------|
| 1               | -1    | 0.4           | -0.6  | -2       | 0.016    | $< 0$                 | revert   |
| 2               | -1    | 0.2           | -0.8  | -2       | -0.728   | $> 0$                 | continue |
| 3               | -0.8  | 0.2           | -0.6  | -0.728   | 0.016    | $< 0$                 | revert   |
| 4               | -0.8  | 0.1           | -0.7  | -0.728   | -0.302   | $> 0$                 | continue |
| 5               | -0.7  | 0.1           | -0.6  | -0.302   | 0.016    | $< 0$                 | revert   |
| 6               | -0.7  | 0.05          | -0.65 | -0.302   | -0.131   | $> 0$                 | continue |
| 7               | -0.65 | 0.05          | -0.6  | -0.131   | 0.016    | $< 0$                 | revert   |
| 8               | -0.65 | 0.01          | -0.64 | -0.131   | -0.09978 |                       |          |



# PAMANTASAN NG LUNGSOD NG SAN PABLO

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| Iteratio<br>ns | $x_1$ | $\triangle x$ | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|----------------|-------|---------------|-------|----------|----------|-----------------------|----------|
| 1              | -1    | 0.4           | -0.6  | -2       | 0.016    | $< 0$                 | revert   |
| 2              | -1    | 0.2           | -0.8  | -2       | -0.728   | $> 0$                 | continue |
| 3              | -0.8  | 0.2           | -0.6  | -0.728   | 0.016    | $< 0$                 | revert   |
| 4              | -0.8  | 0.1           | -0.7  | -0.728   | -0.302   | $> 0$                 | continue |
| 5              | -0.7  | 0.1           | -0.6  | -0.302   | 0.016    | $< 0$                 | revert   |
| 6              | -0.7  | 0.05          | -0.65 | -0.302   | -0.131   | $> 0$                 | continue |
| 7              | -0.65 | 0.05          | -0.6  | -0.131   | 0.016    | $< 0$                 | revert   |
| 8              | -0.65 | 0.01          | -0.64 | -0.131   | -0.09978 | $> 0$                 | continue |

[illegible]

[illegible]

[illegible]







| Iterations | $x_1$ | $\Delta x$ | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|------------|-------|------------|-------|----------|----------|-----------------------|----------|
| 8          | -0.65 | 0.01       | -0.64 | -0.131   | -0.09978 | $> 0$                 | continue |
| 9          | -0.64 | 0.01       | -0.63 | -0.09978 | -0.06949 | $> 0$                 | continue |
| 10         | -0.63 | 0.01       | -0.62 | -0.06949 | -0.04011 |                       |          |
|            |       |            |       |          |          |                       |          |
|            |       |            |       |          |          |                       |          |
|            |       |            |       |          |          |                       |          |
|            |       |            |       |          |          |                       |          |
|            |       |            |       |          |          |                       |          |



| Iterations | $x_1$ | $\Delta x$ | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|------------|-------|------------|-------|----------|----------|-----------------------|----------|
| 8          | -0.65 | 0.01       | -0.64 | -0.131   | -0.09978 | $> 0$                 | continue |
| 9          | -0.64 | 0.01       | -0.63 | -0.09978 | -0.06949 | $> 0$                 | continue |
| 10         | -0.63 | 0.01       | -0.62 | -0.06949 | -0.04011 | $> 0$                 | continue |
|            |       |            |       |          |          |                       |          |
|            |       |            |       |          |          |                       |          |
|            |       |            |       |          |          |                       |          |
|            |       |            |       |          |          |                       |          |
|            |       |            |       |          |          |                       |          |



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| Iterat<br>ions | $x_1$ | $\Delta x$ | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|----------------|-------|------------|-------|----------|----------|-----------------------|----------|
| 8              | -0.65 | 0.01       | -0.64 | -0.131   | -0.09978 | $> 0$                 | continue |
| 9              | -0.64 | 0.01       | -0.63 | -0.09978 | -0.06949 | $> 0$                 | continue |
| 10             | -0.63 | 0.01       | -0.62 | -0.06949 | -0.04011 | $> 0$                 | continue |
| 11             | -0.62 | 0.01       | -0.61 | -0.04011 |          |                       |          |
|                |       |            |       |          |          |                       |          |
|                |       |            |       |          |          |                       |          |
|                |       |            |       |          |          |                       |          |
|                |       |            |       |          |          |                       |          |



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| Iterations | $x_1$ | $\Delta x$ | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|------------|-------|------------|-------|----------|----------|-----------------------|----------|
| 8          | -0.65 | 0.01       | -0.64 | -0.131   | -0.09978 | $> 0$                 | continue |
| 9          | -0.64 | 0.01       | -0.63 | -0.09978 | -0.06949 | $> 0$                 | continue |
| 10         | -0.63 | 0.01       | -0.62 | -0.06949 | -0.04011 | $> 0$                 | continue |
| 11         | -0.62 | 0.01       | -0.61 | -0.04011 | -0.01162 |                       |          |
|            |       |            |       |          |          |                       |          |
|            |       |            |       |          |          |                       |          |
|            |       |            |       |          |          |                       |          |
|            |       |            |       |          |          |                       |          |



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| Iterations | $x_1$ | $\Delta x$ | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|------------|-------|------------|-------|----------|----------|-----------------------|----------|
| 8          | -0.65 | 0.01       | -0.64 | -0.131   | -0.09978 | $> 0$                 | continue |
| 9          | -0.64 | 0.01       | -0.63 | -0.09978 | -0.06949 | $> 0$                 | continue |
| 10         | -0.63 | 0.01       | -0.62 | -0.06949 | -0.04011 | $> 0$                 | continue |
| 11         | -0.62 | 0.01       | -0.61 | -0.04011 | -0.01162 | $> 0$                 | continue |
|            |       |            |       |          |          |                       |          |
|            |       |            |       |          |          |                       |          |
|            |       |            |       |          |          |                       |          |
|            |       |            |       |          |          |                       |          |



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| Iterat<br>ions | $x_1$ | $\Delta x$ | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|----------------|-------|------------|-------|----------|----------|-----------------------|----------|
| 8              | -0.65 | 0.01       | -0.64 | -0.131   | -0.09978 | $> 0$                 | continue |
| 9              | -0.64 | 0.01       | -0.63 | -0.09978 | -0.06949 | $> 0$                 | continue |
| 10             | -0.63 | 0.01       | -0.62 | -0.06949 | -0.04011 | $> 0$                 | continue |
| 11             | -0.62 | 0.01       | -0.61 | -0.04011 | -0.01162 | $> 0$                 | continue |
| 12             | -0.61 | 0.01       | -0.6  | -0.01162 |          |                       |          |
|                |       |            |       |          |          |                       |          |
|                |       |            |       |          |          |                       |          |
|                |       |            |       |          |          |                       |          |



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| Iterat<br>ions | $x_1$ | $\Delta x$ | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|----------------|-------|------------|-------|----------|----------|-----------------------|----------|
| 8              | -0.65 | 0.01       | -0.64 | -0.131   | -0.09978 | $> 0$                 | continue |
| 9              | -0.64 | 0.01       | -0.63 | -0.09978 | -0.06949 | $> 0$                 | continue |
| 10             | -0.63 | 0.01       | -0.62 | -0.06949 | -0.04011 | $> 0$                 | continue |
| 11             | -0.62 | 0.01       | -0.61 | -0.04011 | -0.01162 | $> 0$                 | continue |
| 12             | -0.61 | 0.01       | -0.6  | -0.01162 | 0.016    |                       |          |
|                |       |            |       |          |          |                       |          |
|                |       |            |       |          |          |                       |          |
|                |       |            |       |          |          |                       |          |





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| Iterat<br>ions | $x_1$ | $\Delta x$ | $x_n$ | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|----------------|-------|------------|-------|----------|----------|-----------------------|----------|
| 8              | -0.65 | 0.01       | -0.64 | -0.131   | -0.09978 | $> 0$                 | continue |
| 9              | -0.64 | 0.01       | -0.63 | -0.09978 | -0.06949 | $> 0$                 | continue |
| 10             | -0.63 | 0.01       | -0.62 | -0.06949 | -0.04011 | $> 0$                 | continue |
| 11             | -0.62 | 0.01       | -0.61 | -0.04011 | -0.01162 | $> 0$                 | continue |
| 12             | -0.61 | 0.01       | -0.6  | -0.01162 | 0.016    | $< 0$                 | revert   |
|                |       |            |       |          |          |                       |          |
|                |       |            |       |          |          |                       |          |
|                |       |            |       |          |          |                       |          |



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| Iterat<br>ions | $x_1$ | $\Delta x$ | $x_n$  | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|----------------|-------|------------|--------|----------|----------|-----------------------|----------|
| 8              | -0.65 | 0.01       | -0.64  | -0.131   | -0.09978 | $> 0$                 | continue |
| 9              | -0.64 | 0.01       | -0.63  | -0.09978 | -0.06949 | $> 0$                 | continue |
| 10             | -0.63 | 0.01       | -0.62  | -0.06949 | -0.04011 | $> 0$                 | continue |
| 11             | -0.62 | 0.01       | -0.61  | -0.04011 | -0.01162 | $> 0$                 | continue |
| 12             | -0.61 | 0.01       | -0.6   | -0.01162 | 0.016    | $< 0$                 | revert   |
| 13             | -0.61 | 0.001      | -0.609 | -0.01162 |          |                       |          |
|                |       |            |        |          |          |                       |          |
|                |       |            |        |          |          |                       |          |



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| Iterat<br>ions | $x_1$ | $\Delta x$ | $x_n$  | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|----------------|-------|------------|--------|----------|----------|-----------------------|----------|
| 8              | -0.65 | 0.01       | -0.64  | -0.131   | -0.09978 | $> 0$                 | continue |
| 9              | -0.64 | 0.01       | -0.63  | -0.09978 | -0.06949 | $> 0$                 | continue |
| 10             | -0.63 | 0.01       | -0.62  | -0.06949 | -0.04011 | $> 0$                 | continue |
| 11             | -0.62 | 0.01       | -0.61  | -0.04011 | -0.01162 | $> 0$                 | continue |
| 12             | -0.61 | 0.01       | -0.6   | -0.01162 | 0.016    | $< 0$                 | revert   |
| 13             | -0.61 | 0.001      | -0.609 | -0.01162 | -0.00882 |                       |          |
|                |       |            |        |          |          |                       |          |
|                |       |            |        |          |          |                       |          |



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| Iterat<br>ions | $x_1$ | $\Delta x$ | $x_n$  | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|----------------|-------|------------|--------|----------|----------|-----------------------|----------|
| 8              | -0.65 | 0.01       | -0.64  | -0.131   | -0.09978 | $> 0$                 | continue |
| 9              | -0.64 | 0.01       | -0.63  | -0.09978 | -0.06949 | $> 0$                 | continue |
| 10             | -0.63 | 0.01       | -0.62  | -0.06949 | -0.04011 | $> 0$                 | continue |
| 11             | -0.62 | 0.01       | -0.61  | -0.04011 | -0.01162 | $> 0$                 | continue |
| 12             | -0.61 | 0.01       | -0.6   | -0.01162 | 0.016    | $< 0$                 | revert   |
| 13             | -0.61 | 0.001      | -0.609 | -0.01162 | -0.00882 | $> 0$                 | continue |
|                |       |            |        |          |          |                       |          |
|                |       |            |        |          |          |                       |          |



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| Iterat<br>ions | $x_1$  | $\Delta x$ | $x_n$  | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|----------------|--------|------------|--------|----------|----------|-----------------------|----------|
| 8              | -0.65  | 0.01       | -0.64  | -0.131   | -0.09978 | $> 0$                 | continue |
| 9              | -0.64  | 0.01       | -0.63  | -0.09978 | -0.06949 | $> 0$                 | continue |
| 10             | -0.63  | 0.01       | -0.62  | -0.06949 | -0.04011 | $> 0$                 | continue |
| 11             | -0.62  | 0.01       | -0.61  | -0.04011 | -0.01162 | $> 0$                 | continue |
| 12             | -0.61  | 0.01       | -0.6   | -0.01162 | 0.016    | $< 0$                 | revert   |
| 13             | -0.61  | 0.001      | -0.609 | -0.01162 | -0.00882 | $> 0$                 | continue |
| 14             | -0.609 | 0.001      | -0.608 | -0.00882 |          |                       |          |
|                |        |            |        |          |          |                       |          |



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| Iterat<br>ions | $x_1$  | $\Delta x$ | $x_n$  | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|----------------|--------|------------|--------|----------|----------|-----------------------|----------|
| 8              | -0.65  | 0.01       | -0.64  | -0.131   | -0.09978 | $> 0$                 | continue |
| 9              | -0.64  | 0.01       | -0.63  | -0.09978 | -0.06949 | $> 0$                 | continue |
| 10             | -0.63  | 0.01       | -0.62  | -0.06949 | -0.04011 | $> 0$                 | continue |
| 11             | -0.62  | 0.01       | -0.61  | -0.04011 | -0.01162 | $> 0$                 | continue |
| 12             | -0.61  | 0.01       | -0.6   | -0.01162 | 0.016    | $< 0$                 | revert   |
| 13             | -0.61  | 0.001      | -0.609 | -0.01162 | -0.00882 | $> 0$                 | continue |
| 14             | -0.609 | 0.001      | -0.608 | -0.00882 | -0.00603 |                       |          |
|                |        |            |        |          |          |                       |          |



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| Iterat<br>ions | $x_1$  | $\Delta x$ | $x_n$  | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|----------------|--------|------------|--------|----------|----------|-----------------------|----------|
| 8              | -0.65  | 0.01       | -0.64  | -0.131   | -0.09978 | $> 0$                 | continue |
| 9              | -0.64  | 0.01       | -0.63  | -0.09978 | -0.06949 | $> 0$                 | continue |
| 10             | -0.63  | 0.01       | -0.62  | -0.06949 | -0.04011 | $> 0$                 | continue |
| 11             | -0.62  | 0.01       | -0.61  | -0.04011 | -0.01162 | $> 0$                 | continue |
| 12             | -0.61  | 0.01       | -0.6   | -0.01162 | 0.016    | $< 0$                 | revert   |
| 13             | -0.61  | 0.001      | -0.609 | -0.01162 | -0.00882 | $> 0$                 | continue |
| 14             | -0.609 | 0.001      | -0.608 | -0.00882 | -0.00603 | $> 0$                 | continue |
|                |        |            |        |          |          |                       |          |



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| Iterat<br>ions | $x_1$  | $\Delta x$ | $x_n$  | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|----------------|--------|------------|--------|----------|----------|-----------------------|----------|
| 8              | -0.65  | 0.01       | -0.64  | -0.131   | -0.09978 | $> 0$                 | continue |
| 9              | -0.64  | 0.01       | -0.63  | -0.09978 | -0.06949 | $> 0$                 | continue |
| 10             | -0.63  | 0.01       | -0.62  | -0.06949 | -0.04011 | $> 0$                 | continue |
| 11             | -0.62  | 0.01       | -0.61  | -0.04011 | -0.01162 | $> 0$                 | continue |
| 12             | -0.61  | 0.01       | -0.6   | -0.01162 | 0.016    | $< 0$                 | revert   |
| 13             | -0.61  | 0.001      | -0.609 | -0.01162 | -0.00882 | $> 0$                 | continue |
| 14             | -0.609 | 0.001      | -0.608 | -0.00882 | -0.00603 | $> 0$                 | continue |
| 15             | -0.608 | 0.001      | -0.607 | -0.00603 |          |                       |          |





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| Iterat<br>ions | $x_1$  | $\Delta x$ | $x_n$  | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|----------------|--------|------------|--------|----------|----------|-----------------------|----------|
| 8              | -0.65  | 0.01       | -0.64  | -0.131   | -0.09978 | $> 0$                 | continue |
| 9              | -0.64  | 0.01       | -0.63  | -0.09978 | -0.06949 | $> 0$                 | continue |
| 10             | -0.63  | 0.01       | -0.62  | -0.06949 | -0.04011 | $> 0$                 | continue |
| 11             | -0.62  | 0.01       | -0.61  | -0.04011 | -0.01162 | $> 0$                 | continue |
| 12             | -0.61  | 0.01       | -0.6   | -0.01162 | 0.016    | $< 0$                 | revert   |
| 13             | -0.61  | 0.001      | -0.609 | -0.01162 | -0.00882 | $> 0$                 | continue |
| 14             | -0.609 | 0.001      | -0.608 | -0.00882 | -0.00603 | $> 0$                 | continue |
| 15             | -0.608 | 0.001      | -0.607 | -0.00603 | -0.00325 |                       |          |



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| Iterat<br>ions | $x_1$  | $\Delta x$ | $x_n$  | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|----------------|--------|------------|--------|----------|----------|-----------------------|----------|
| 8              | -0.65  | 0.01       | -0.64  | -0.131   | -0.09978 | $> 0$                 | continue |
| 9              | -0.64  | 0.01       | -0.63  | -0.09978 | -0.06949 | $> 0$                 | continue |
| 10             | -0.63  | 0.01       | -0.62  | -0.06949 | -0.04011 | $> 0$                 | continue |
| 11             | -0.62  | 0.01       | -0.61  | -0.04011 | -0.01162 | $> 0$                 | continue |
| 12             | -0.61  | 0.01       | -0.6   | -0.01162 | 0.016    | $< 0$                 | revert   |
| 13             | -0.61  | 0.001      | -0.609 | -0.01162 | -0.00882 | $> 0$                 | continue |
| 14             | -0.609 | 0.001      | -0.608 | -0.00882 | -0.00603 | $> 0$                 | continue |
| 15             | -0.608 | 0.001      | -0.607 | -0.00603 | -0.00325 | $> 0$                 | continue |



# PAMANTASAN NG LUNGSOD NG SAN PABLO

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| Iterations | $x_1$  | $\Delta x$ | $x_n$  | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|------------|--------|------------|--------|----------|----------|-----------------------|----------|
| 15         | -0.608 | 0.001      | -0.607 | -0.00603 | -0.00325 | $> 0$                 | continue |
| 16         | -0.607 | 0.001      | -0.606 | -0.00325 |          |                       |          |



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| Iterations | $x_1$  | $\Delta x$ | $x_n$  | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|------------|--------|------------|--------|----------|----------|-----------------------|----------|
| 15         | -0.608 | 0.001      | -0.607 | -0.00603 | -0.00325 | $> 0$                 | continue |
| 16         | -0.607 | 0.001      | -0.606 | -0.00325 | -0.00047 | $> 0$                 |          |



# PAMANTASAN NG LUNGSOD NG SAN PABLO

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| Iterations | $x_1$  | $\Delta x$ | $x_n$  | $f(x_1)$ | $f(x_n)$ | $f(x_1) \cdot f(x_n)$ | Remarks  |
|------------|--------|------------|--------|----------|----------|-----------------------|----------|
| 15         | -0.608 | 0.001      | -0.607 | -0.00603 | -0.00325 | $> 0$                 | continue |
| 16         | -0.607 | 0.001      | -0.606 | -0.00325 | -0.00047 | $> 0$                 |          |

The approximate root of  $f(x) = 4x^3 + 3x^2 + 2x + 1$  is -0.606.



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# ACTIVITY # 3



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1. Use the incremental-search method with  $\Delta x = 0.2$  to bracket the smallest positive zero of  $f(x) = x^3 - 10x^2 + 5$  with  $x_1 = 0$ .
2. Use the incremental-search method of  $f(x) = e^x - 3$ , with  $x_1 = 1$  and  $\Delta x = 0.3$ .



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